

P3 Advanced practical skills

Practical · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Use safe measured quantities and observations to make reproducible chemical inferences. In titration rinse burette with titrant, pipette with analyte and repeat to concordance; record readings with appropriate decimal places. Qualitative tests need reagent, observation and inference, plus alternatives where observations overlap.

Worked example: follow the reasoning

For a 25.0 cm^3 sample and mean titre 20.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ acid reacting 1:1, sample contains 0.00200 mol base; concentration is $0.0800 \text{ mol dm}^{-3}$.

Quick check · multiple choice

A Which good titration practice improves reliability?

[1]

- A One rough titre only
- B Repeat to concordance
- C Read volume to nearest litre
- D Rinse burette with analyte

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A mean 25.0 cm^3 titre of $0.100 \text{ mol dm}^{-3}$ HCl neutralises 25.0 cm^3 NaOH. Determine NaOH concentration and state one titration good practice.

[3]

Final answer: _____

Answers and explanation

Quick check: B. Repeat to concordance

Question 1 [3 marks]

$n(\text{HCl}) = 0.00250 \text{ mol}$ [1]; 1:1 gives $[\text{NaOH}] = 0.100 \text{ mol dm}^{-3}$ [1]; use concordant titres/read burette to two decimal places [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.